Microbial growth and nutrition

Nutrient Requirements

- Energy Source
  - Phototroph
    - Uses light as an energy source
  - Chemotroph
    - Uses energy from the oxidation of reduced chemical compounds

Nutrient Requirements

- Electron (Reduction potential) Source
  - Organotroph
    - Uses reduced organic compounds as a source for reduction potential
  - Lithotroph
    - Uses reduced inorganic compounds as a source for reduction potential

Nutrient Requirements

- Carbon source
  - Autotroph
    - Can use CO_2 as a sole carbon source (Carbon fixation)
  - Heterotroph
    - Requires an organic carbon source; cannot use CO_2 as a carbon source

Nutrient Requirements

- Nitrogen source
  - Organic nitrogen
    - Primarily from the catabolism of amino acids
  - Oxidized forms of inorganic nitrogen
    - Nitrate (NO_3^{2-}) and nitrite (NO_2^-)
  - Reduced inorganic nitrogen
    - Ammonium (NH_4^+)
  - Dissolved nitrogen gas (N_2) (Nitrogen fixation)

Nutrient Requirements

- Phosphate source
  - Organic phosphate
  - Inorganic phosphate ($H_2PO_4^-$ and HPO_4^{2-})

Nutrient Requirements

- Sulfur source
  - Organic sulfur
  - Oxidized inorganic sulfur
    - Sulfate (SO_4^{2-})
  - Reduced inorganic sulfur
    - Sulfide (S^{2-} or H_2S)
  - Elemental sulfur (S_o)

Nutrient Requirements

- Special requirements
  - Amino acids
  - Nucleotide bases
  - Enzymatic cofactors or “vitamins”

Nutrient Requirements

- Prototrophs vs. Auxotrophs
  - Prototroph
    - A species or genetic strain of microbe capable of growing on a minimal medium consisting a simple carbohydrate or CO_2 carbon source, with inorganic sources of all other nutrient requirements
  - Auxotroph
    - A species or genetic strain requiring one or more complex organic nutrients (such as amino acids, nucleotide bases, or enzymatic cofactors) for growth

Nutrient Transport Processes

- Simple Diffusion
  - Movement of substances directly across a phospholipid bilayer, with no need for a transport protein
  - Movement from high $\rightarrow$ low concentration
  - No energy expenditure (e.g. ATP) from cell
  - Small uncharged molecules may be transported via this process, e.g. H_2O, O_2, CO_2

Nutrient Transport Processes

- Facilitated Diffusion
  - Movement of substances across a membrane with the assistance of a transport protein
  - Movement from high $\rightarrow$ low concentration
  - No energy expenditure (e.g. ATP) from cell
  - Two mechanisms: Channel & Carrier Proteins

Nutrient Transport Processes

- Active Transport
  - Movement of substances across a membrane with the assistance of a transport protein
  - Movement from low $\rightarrow$ high concentration
  - Energy expenditure (e.g. [ATP] or [ion gradients]) from cell
  - Active transport pumps are usually carrier proteins

Nutrient Transport Processes

- Active Transport (cont.)
  - Active transport systems in bacteria
    - **ATP-binding cassette transporters (ABC transporters):**
      The target binds to a soluble cassette protein (in periplasm of gram-negative bacterium, or located bound to outer leaflet of plasma membrane in gram-positive bacterium). The target-cassette complex then binds to an integral membrane ATPase pump that transports the target across the plasma membrane.

Nutrient Transport Processes

- Active Transport (cont.)
  - Active transport systems in bacteria
    - **Cotransport systems:** Transport of one substance from a low → high concentration as another substance is simultaneously transported from high → low.

      For example: lactose permease in *E. coli:*
      As hydrogen ions are moved from a high concentration outside → low concentration inside, lactose is moved from a low concentration outside → high concentration inside

Nutrient Transport Processes

- Active Transport (cont.)
  - Active transport systems in bacteria
    - **Group translocation system:** A molecule is transported while being chemically modified.

      For example:
      phosphoenolpyruvate: sugar phosphotransferase systems (PTS)

      PEP + sugar (outside) $\rightarrow$ pyruvate + sugar-phosphate (inside)

Nutrient Transport Processes

- Active Transport (cont.)
  - Active transport systems in bacteria
    - **Iron uptake by siderophores**:

      Low molecular weight organic molecules that are secreted by bacteria to bind to ferric iron (Fe^{3+}); necessary due to low solubility of iron; Fe^{3+}-siderophore complex is then transported via ABC transporter

Microbiological Media

- Liquid (broth) vs. semisolid media
  - Liquid medium
    - Components are dissolved in water and sterilized
  - Semisolid medium
    - A medium to which has been added a gelling agent
    - Agar (most commonly used)
    - Gelatin
    - Silica gel (used when a non-organic gelling agent is required)

Microbiological Media

- Chemically defined vs. complex media
  - Chemically defined media
    - The exact chemical composition is known
    - e.g. minimal media used in bacterial genetics experiments
  - Complex media
    - Exact chemical composition is not known
    - Often consist of plant or animal extracts, such as soybean meal, milk protein, etc.
    - Include most routine laboratory media, e.g., tryptic soy broth

Microbiological Media

- Selective media
  - Contain agents that inhibit the growth of certain bacteria while permitting the growth of others
  - Frequently used to isolate specific organisms from a large population of contaminants
- Differential media
  - Contain indicators that react differently with different organisms (for example, producing colonies with different colors)
  - Used in identifying specific organisms

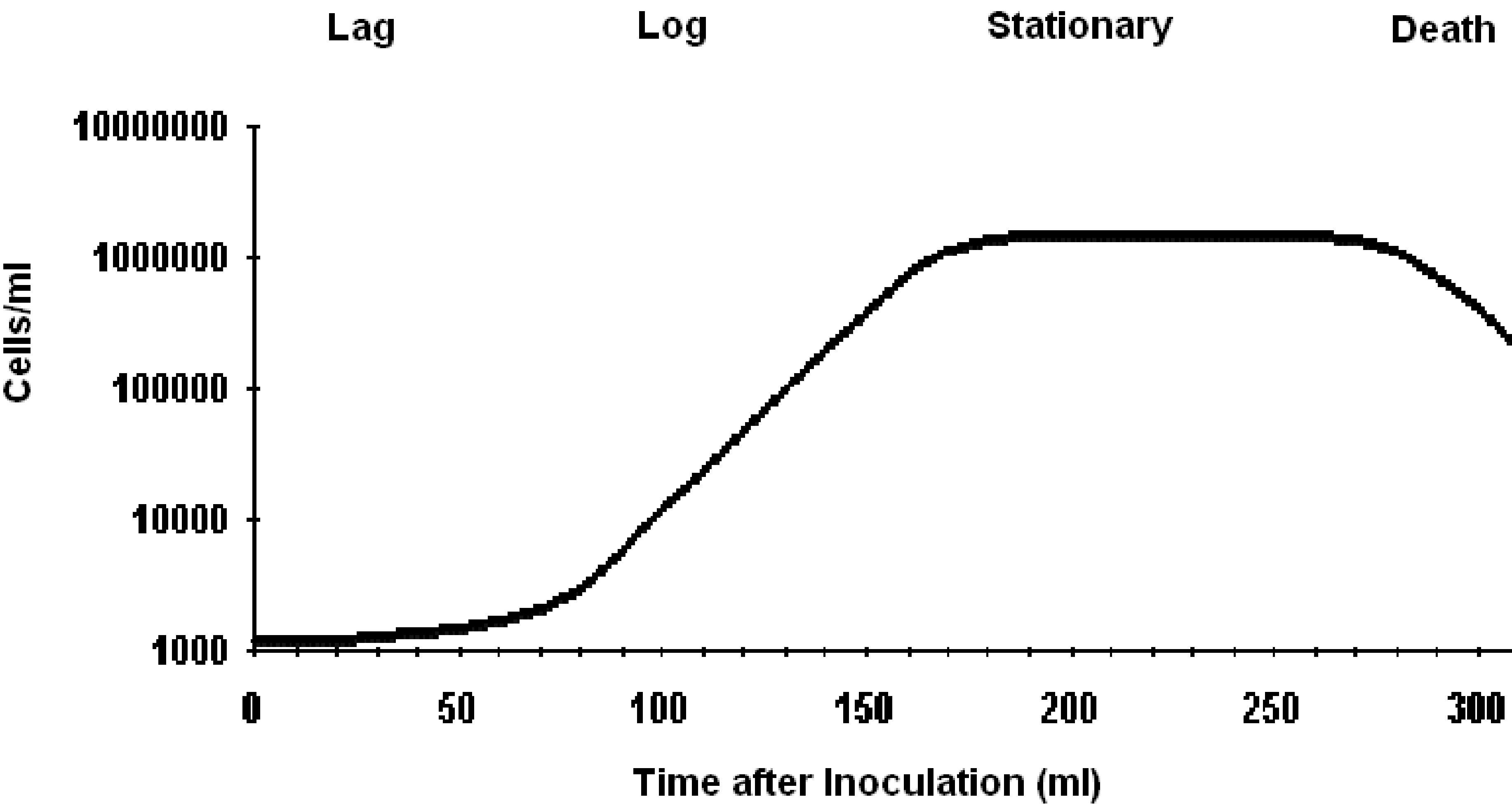
Growth in Batch Culture

- “Growth” is generally used to refer to the acquisition of biomass leading to cell division, or reproduction
- A “batch culture” is a closed system in broth medium in which no additional nutrient is added after inoculation of the broth.

Growth in Batch Culture

- Typically, a batch culture passes through four distinct stages:
  - Lag stage
  - Logarithmic (exponential) growth
  - Stationary stage
  - Death stage

Growth in Batch Culture



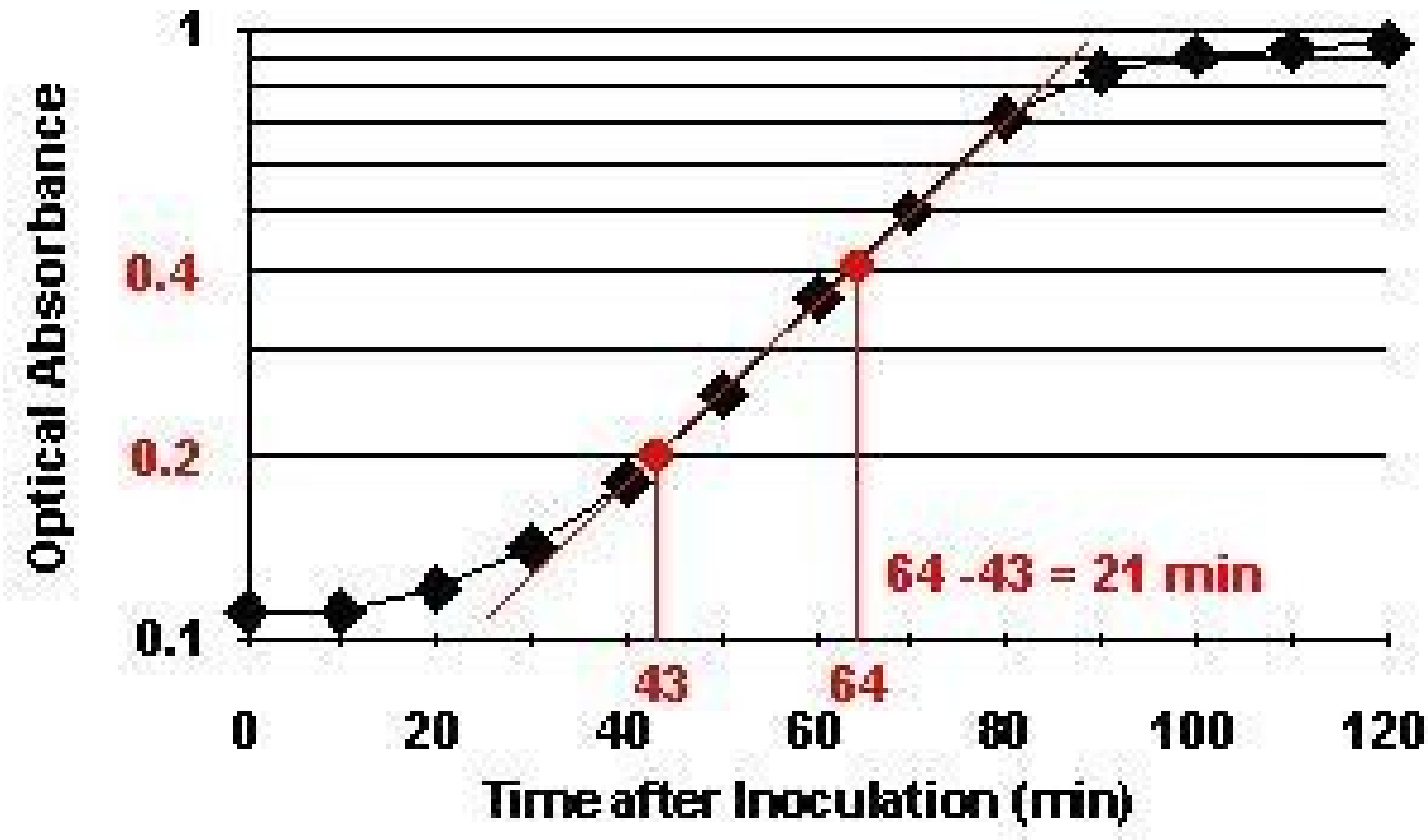
Mean Generation Time and Growth Rate

- The mean generation time (doubling time) is the amount of time required for the concentration of cells to double during the log stage. It is expressed in units of minutes.

- Growth rate (min^{-1}) = $$\frac{1}{\text{mean generation time}}$$

- Mean generation time can be determined directly from a semilog plot of bacterial concentration vs time after inoculation

Mean Generation Time and Growth Rate

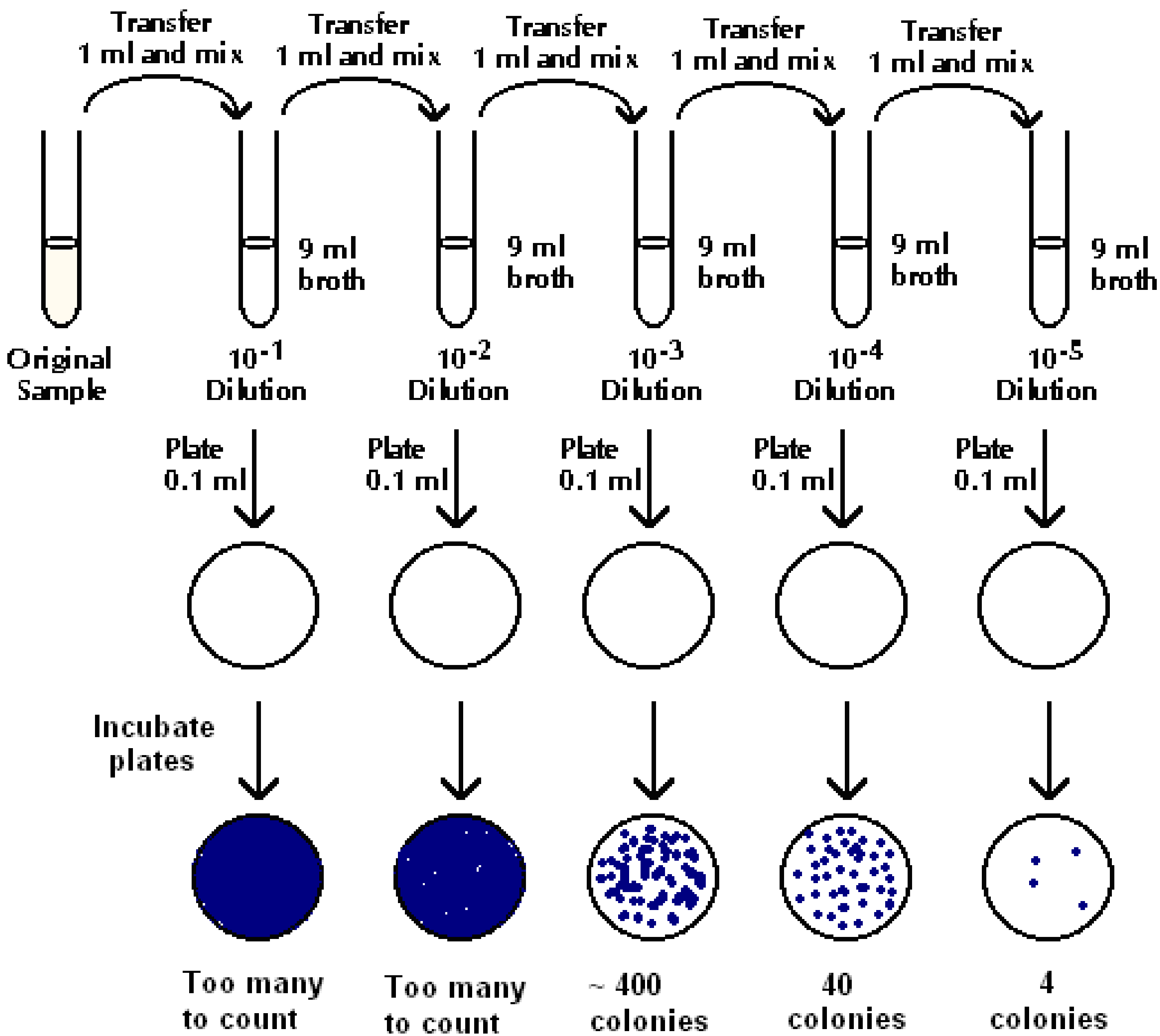


Measurement of Microbial Growth

- Microscopic cell counts
  - Calibrated "Petroff-Hausser counting chamber," similar to hemacytometer, can be used
  - Generally very difficult for bacteria since cells tend to move in and out of counting field
  - Can be useful for organisms that can't be cultured
  - Special stains (e.g. serological stains or stains for viable cells) can be used for specific purposes
- Serial dilution and colony counting
  - Also know as "viable cell counts"
  - Concentrated samples are diluted by serial dilution

Measurement of Microbial Growth

- Serial dilution and colony counting
  - Also know as “viable cell counts”
  - Concentrated samples are diluted by serial dilution
  - The diluted samples can be either plated by spread plating or by pour plating


Transfer
1 ml and mix
Transfer
1 ml and mix
Transfer
1 ml and mix
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1 ml and mix
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1 ml and mix
9 ml
broth
9 ml
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9 ml
broth
9 ml
broth
9 ml
broth
Original
Sample
10^{-1}
Dilution
10^{-2}
Dilution
10^{-3}
Dilution
10^{-4}
Dilution
10^{-5}
Dilution
Plate
0.1 ml
Plate
0.1 ml
Plate
0.1 ml
Plate
0.1 ml
Plate
0.1 ml
Incubate
plates
Too many
to count
Too many
to count
~ 400
colonies
40
colonies
4
colonies

Measurement of Microbial Growth

- Serial dilution (cont.)
  - Diluted samples are spread onto media in petri dishes and incubated
  - Colonies are counted. The concentration of bacteria in the original sample is calculated (from plates with 25 – 250 colonies, from the *FDA Bacteriological Analytical Manual*).
  - A simple calculation, with a single plate falling into the statistically valid range, is given below:

$$\frac{\text{CFU}}{\text{ml}} \text{ in original sample} = \frac{\text{\# colonies counted}}{(\text{dilution factor})(\text{volume plated, in ml})}$$

Measurement of Microbial Growth

- Serial dilution (cont.)
  - If there is **more than one plate** in the statistically valid range of 25 – 250 colonies, the viable cell count is determined by the following formula:

Measurement of Microbial Growth

$$\frac{\text{CFU}}{\text{ml}} = \frac{\sum C}{[(1 * n_1) + (0.1 * n_2) + \ldots] * d_1 * V}$$

- Where:
  C = Sum of all colonies on all plates between 25 - 250
  n_1= number of plates counted at dilution 1
  (least diluted plate counted)
  n_2= number of plates counted at dilution 2
  (dilution 2 = 0.1 of dilution 1)
  d_1= dilution factor of dilution 1
  V= Volume plated per plate

Measurement of Microbial Growth

- Membrane filtration
  - Used for samples with low microbial concentration
  - A measured volume (usually 1 to 100 ml) of sample is filtered through a membrane filter (typically with a 0.45 µm pore size)
  - The filter is placed on a nutrient agar medium and incubated
  - Colonies grow on the filter and can be counted

Measurement of Microbial Growth

- Turbidity
  - Based on the diffraction or “scattering” of light by bacteria in a broth culture
  - Light scattering is measured as optical absorbance in a spectrophotometer
  - Optical absorbance is directly proportional to the concentration of bacteria in the suspension

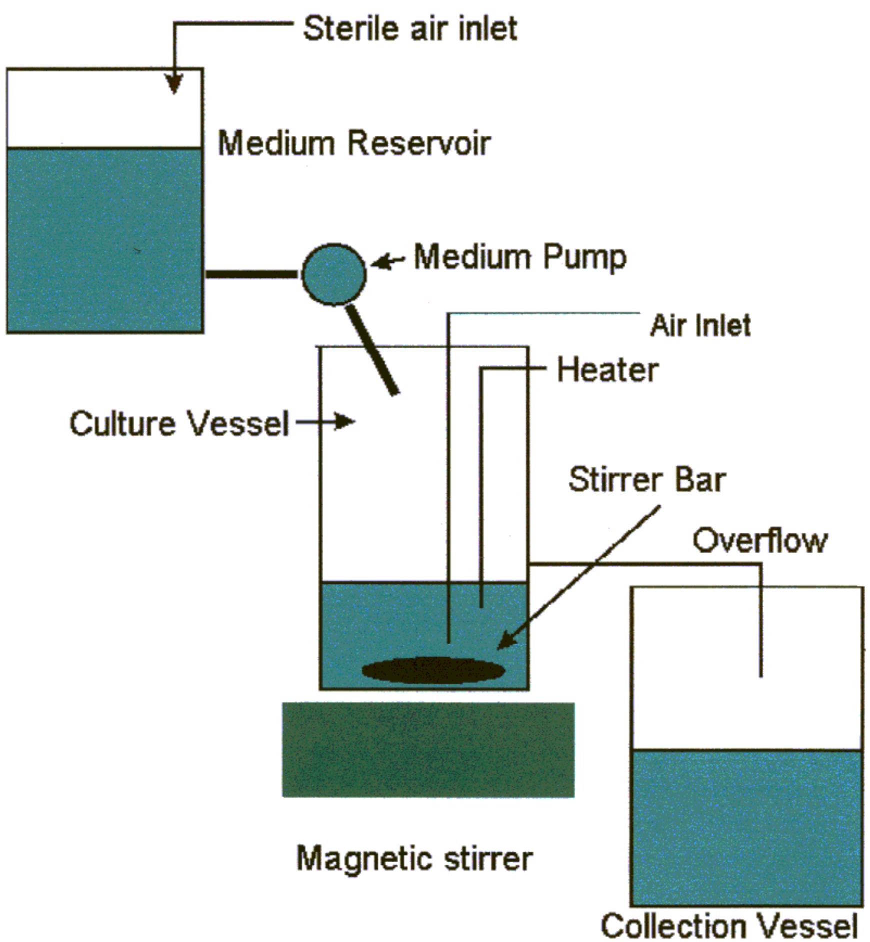
Measurement of Microbial Growth

- Mass determination
  - Cells are removed from a broth culture by centrifugation and weighed to determine the “wet mass.”
  - The cells can be dried out and weighed to determine the “dry mass.”
- Measurement of enzymatic activity or other cell components

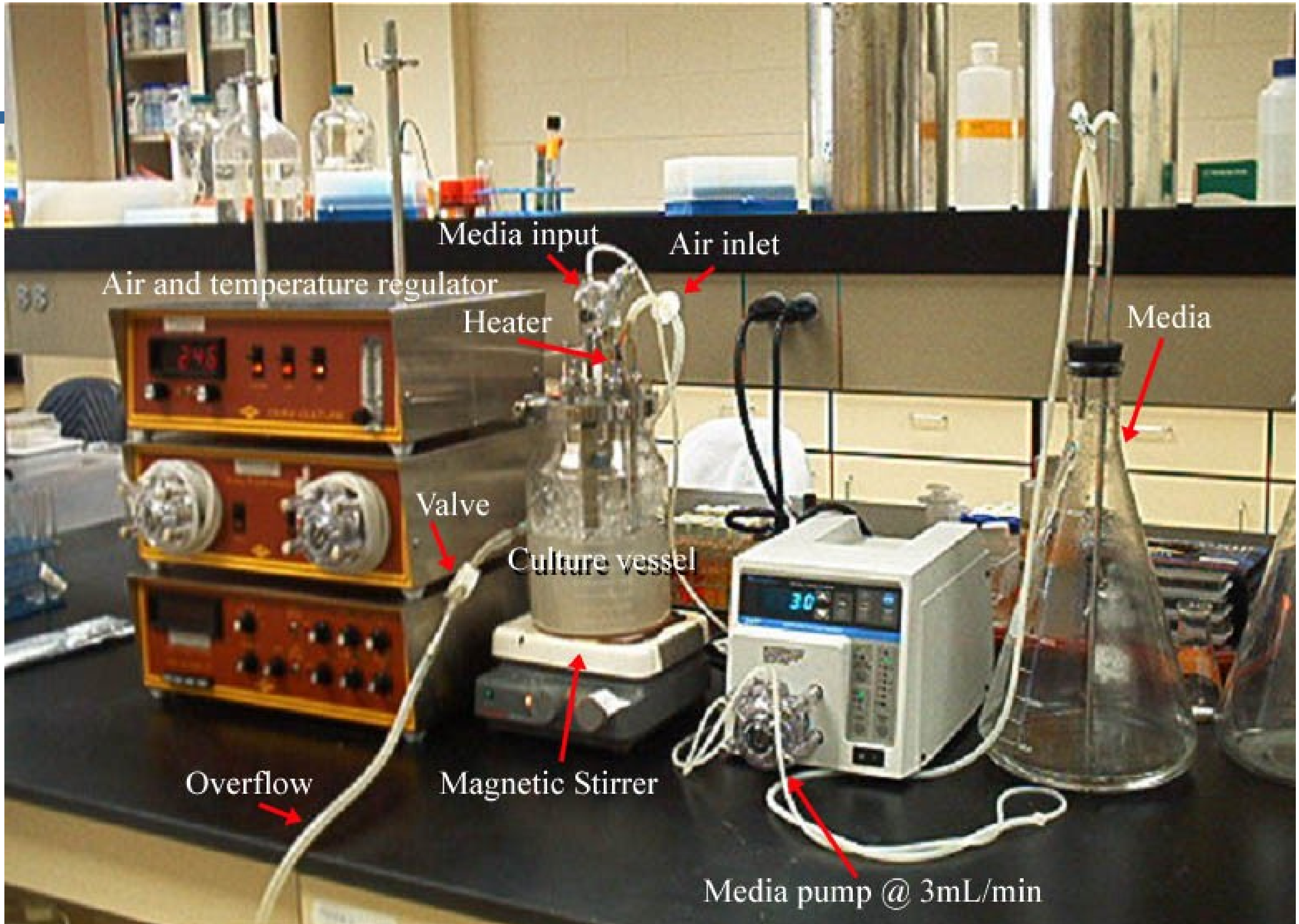
Growth in Continuous Culture

- A “continuous culture” is an open system in which fresh media is continuously added to the culture at a constant rate, and old broth is removed at the same rate.
- This method is accomplished in a device called a chemostat.
- Typically, the concentration of cells will reach an equilibrium level that remains constant as long as the nutrient feed is maintained.

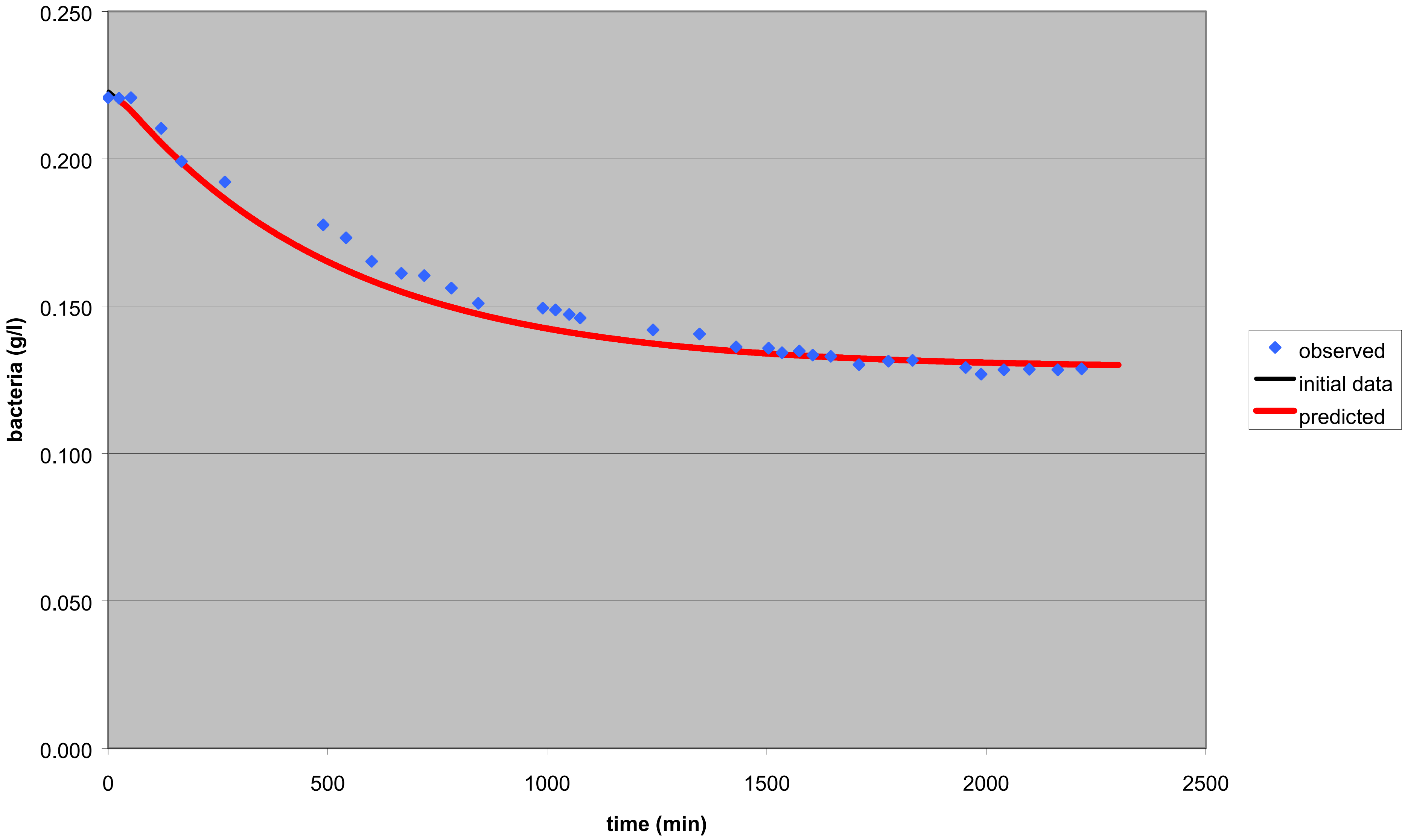
Basic Chemostat System



Our Chemostat System



0.05% glucose run assuming delay of 20 min and 28% yield



Factors that Influence Growth

- Growth vs. Tolerance
  - “Growth” is generally used to refer to the acquisition of biomass leading to cell division, or reproduction
  - Many microbes can survive under conditions in which they cannot grow
  - The suffix “-phile” is often used to describe conditions permitting growth, whereas the term “tolerant” describes conditions in which the organisms survive, but don’t necessarily grow
  - For example, a “thermophilic bacterium” grows under conditions of elevated temperature, while a “thermotolerant bacterium” survives elevated temperature, but grows at a lower temperature

Factors that Influence Growth

- Obligate (strict) vs. facultative
  - “Obligate” (or “strict”) means that a given condition is required for growth
  - “Facultative” means that the organism can grow under the condition, but doesn’t require it
  - The term “facultative” is often applied to sub-optimal conditions
  - For example, an obligate thermophile requires elevated temperatures for growth, while a facultative thermophile may grow in either elevated temperatures or lower temperatures

Factors that Influence Growth

- Temperature
  - Most bacteria grow throughout a range of approximately 20 Celsius degrees, with the maximum growth rate at a certain "optimum temperature"
  - Psychrophiles: Grows well at 0ºC; optimally between 0ºC – 15ºC
  - Psychrotrophs: Can grow at 0 – 10ºC; optimum between 20 – 30ºC and maximum around 35ºC
  - Mesophiles: Optimum around 20 – 45ºC
  - Moderate thermophiles: Optimum around 55 – 65 ºC
  - Extreme thermophiles (Hyperthermophiles): Optimum around 80 – 113 ºC

Factors that Influence Growth

- pH
  - Acidophiles:
    - Grow optimally between ~pH 0 and 5.5
  - Neutrophiles
    - Growoptimally between pH 5.5 and 8
  - Alkalophiles
    - Grow optimally between pH 8 – 11.5

Factors that Influence Growth

- Salt concentration
  - Halophiles require elevated salt concentrations to grow; often require 0.2 M ionic strength or greater and may some may grow at 1 M or greater; example, *Halobacterium*
  - Osmotolerant (halotolerant) organisms grow over a wide range of salt concentrations or ionic strengths; for example, *Staphylococcus aureus*

Factors that Influence Growth

- Oxygen concentration
  - Strict aerobes: Require oxygen for growth (~20%)
  - Strict anaerobes: Grow in the absence of oxygen; cannot grow in the presence of oxygen
  - Facultative anaerobes: Grow best in the presence of oxygen, but are able to grow (at reduced rates) in the absence of oxygen
  - Aerotolerant anaerobes: Can grow equally well in the presence or absence of oxygen
  - Microaerophiles: Require reduced concentrations of oxygen (~2 – 10%) for growth

Quorum Sensing

- A mechanism by which members of a bacterial population can behave cooperatively, altering their patterns of gene expression (transcription) in response to the density of the population
- In this way, the entire population can respond in a manner most strategically practical depending on how sparse or dense the population is.

Quorum Sensing

- Mechanism:
  - As the bacteria in the population grow, they secrete a quorum signaling molecule into the environment (for example, in many gram-negative bacteria the signal is an acyl homoserine lactone, HSL)
  - When the quorum signal reaches a high enough concentration, it triggers specific receptor proteins that usually act as transcriptional inducers, turning on quorum-sensitive genes